Cited Sources in LLM Assistants: Diversity, Credibility, and Coverage Gaps

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Abstract

Large Language Model (LLM) assistants are increasingly used as conversational information-access systems, yet little is known about the sources they cite or how source selection varies across systems. This paper analyzes the citation practices of seven LLM-based assistants using 51 real-world questions about electric vehicles (EVs), a technically complex and policy-relevant domain shaped by commercial interests and polarized public debate. The resulting dataset comprises 1,179 assistant-specific URL records from 357 responses, corresponding to 906 distinct URL hostnames. We assessed these sources using reliability and political-bias ratings from NewsGuard and Media Bias/Fact Check (MBFC), complemented by a qualitative classification of unrated websites. The results show differences across assistants in the number and composition of their aggregated cited-source sets. Most rated URL records are associated with websites receiving favorable credibility assessments. Within the service-specific unrated subsets, business and commercial websites account for the largest average share of manually classified URL records. The cited sources extend beyond mainstream news outlets to an ecosystem only partly covered by existing media-rating frameworks. These findings highlight the need for broader source-assessment approaches in conversational information-access systems.

Keywords

Electric Vehicles, LLMs, Credibility of Cited Sources

1. Introduction

In recent years, Large Language Models (LLMs) have reshaped how citizens access and interpret information. Through conversational interfaces, they increasingly complement and, in some contexts, replace traditional search engines by providing synthesized answers to complex queries [1]. Yet the mechanisms through which LLMs retrieve, prioritize, and integrate information remain under active investigation, particularly with regard to the criteria guiding source selection and ranking [2, 3]. Moreover, the sources displayed in an answer do not necessarily provide a complete account of the information used during its generation. Examining citations therefore cannot reveal the entire generation process, but it provides an observable basis for characterizing the information environment that these systems present to users.

This limited transparency has direct implications for information quality. If LLM-generated responses cite sources that provide incomplete or biased information, users may encounter distorted or misleading representations of reality. When source reliability has not been assessed, evaluating this risk becomes more difficult.

To investigate this issue, we focus on electric vehicles (EVs). The transportation sector is undergoing a structural transition toward electric mobility, a key component of climate mitigation and decarbonization strategies [4]. At the same time, EVs are the object of intense public debate, where technological complexity intersects with policy choices and consumer behavior. This makes the domain particularly sensitive to both misinformation and selective framing.

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The convergence of AI-mediated information access and a contested technological transition makes EVs a useful case for studying source selection in LLM-generated answers. Misleading information may affect perceptions of environmental benefits, safety, economic costs, and technological maturity, influencing individual decisions and public support for related policies [5]. Moreover, documented disinformation campaigns have sought to discredit electric mobility by amplifying concerns about range, safety, performance in adverse weather, and fire risk, often selectively interpreting scientific evidence to foster distrust [6]. Misleading EV-related content is not limited to demonstrably false statements; it may also include selective framing, misleading simplifications, and unsupported assertions [7].

Existing source-assessment services offer one possible basis for examining the information surfaced by LLM-based assistants. NewsGuard provides website-level credibility ratings, while MBFC provides assessments of credibility, factual reporting, and bias.

This paper examines the websites cited by seven LLM-based assistants in response to 51 real-world questions about EVs. Using a common set of prompts, we analyze 1,179 assistant-specific URL records obtained after deduplicating cited URLs within each assistant across its 51 responses. We combine a quantitative comparison of citation patterns with website-level credibility and political-bias ratings from NewsGuard and MBFC. Because a large share of the identified websites falls outside the coverage of these services, we complement the ratings with a qualitative classification of the unrated source ecosystem.

The paper poses three **research questions**:

RQ1. To what extent do LLM-based assistants overlap in the domains they cite across a common set of EV-related questions?

RQ2. Which credibility profiles characterize the sources evaluated by NewsGuard and MBFC?

RQ3. Which types of websites remain outside the coverage of these rating services?

By answering these questions, the paper makes two main **contributions**. First, it provides an empirical, cross-system analysis of citation patterns in conversational answers within a technically complex and socially contested domain. Second, it demonstrates that the source ecosystem surfaced by LLM-based assistants extends beyond the news organizations typically covered by existing credibility-rating frameworks. The findings contribute to the evaluation of LLM-based question-answering and information-access systems and motivate broader approaches to assessing the heterogeneous sources presented to users in AI-mediated information environments.

2. Related Work

Recent research has examined AI search and chat assistants as intermediaries that shape users' exposure to information sources. Yang [8] analyzes over 366,000 citations across multiple AI search systems, finding a strong concentration on a small number of news outlets and a preference for high-quality, predominantly center/left-leaning sources. Li and Sinnamon [9] identify sentiment, commercial, and geographic biases in responses and sources from ChatGPT, Bing Chat, and Perplexity. Kirsten et al. [10] compare Google organic search with generative search systems, reporting differences in source diversity and stability, including variation across time and executions. Together, these studies motivate examining the sources cited by individual assistants while accounting for the variability of their outputs.

A complementary line of research examines source credibility and its relationship to generated answers. Vykopal et al. [11] jointly evaluate source credibility and response groundedness, showing that apparently well-supported answers may still rely on low-credibility or mixed-quality sources. DeVerna et al. [12] show that carefully curated retrieval context, including evidence from recognized fact-checking sources, improves LLM performance in political fact-checking. Their use of NewsGuard and Media Bias/Fact Check also provides a methodological connection to our source-level assessment.

Source credibility is distinct from whether a citation supports the associated claim. Liu et al. [13] audit four generative search engines, distinguishing citation recall, the proportion of generated sentences fully supported by their citations, from citation precision, the proportion of citations supporting their associated sentence. On average, only 51.5% of sentences were fully supported, and 74.5% of citations

supported their associated sentence. Our study addresses the complementary question of which websites assistants cite and how these websites are assessed; it does not evaluate whether citations support the generated answers.

Electric vehicles provide a relevant domain for this analysis. Prior research documents misleading narratives concerning EV fire risk, electromagnetic health effects, and life-cycle emissions [7]. The domain also brings together journalistic, corporate, technical, and industry-specific sources. Although existing studies explore LLMs in EV-related technical applications [14, 15], to the best of our knowledge they do not examine the websites cited by assistants in response to EV-related user questions. Our study addresses this gap by comparing cited-domain sets across seven assistants, examining their coverage and ratings in NewsGuard and MBFC, and characterizing the website types within the service-specific unrated subsets.

3. Pipeline

This section presents the pipeline adopted to investigate and compare the sources referenced by LLMs when generating responses to questions related to EVs. The pipeline includes the collection of EV-related questions from online platforms, the generation of responses from multiple LLMs, the extraction of referenced URLs, and the subsequent analysis of source overlap and source characteristics. Figure 1 provides an overview of the procedure, while Table 2 summarizes the main statistics of the collected dataset. The pipeline can be adapted to other domains, although question collection and source classification may require domain-specific adjustments.

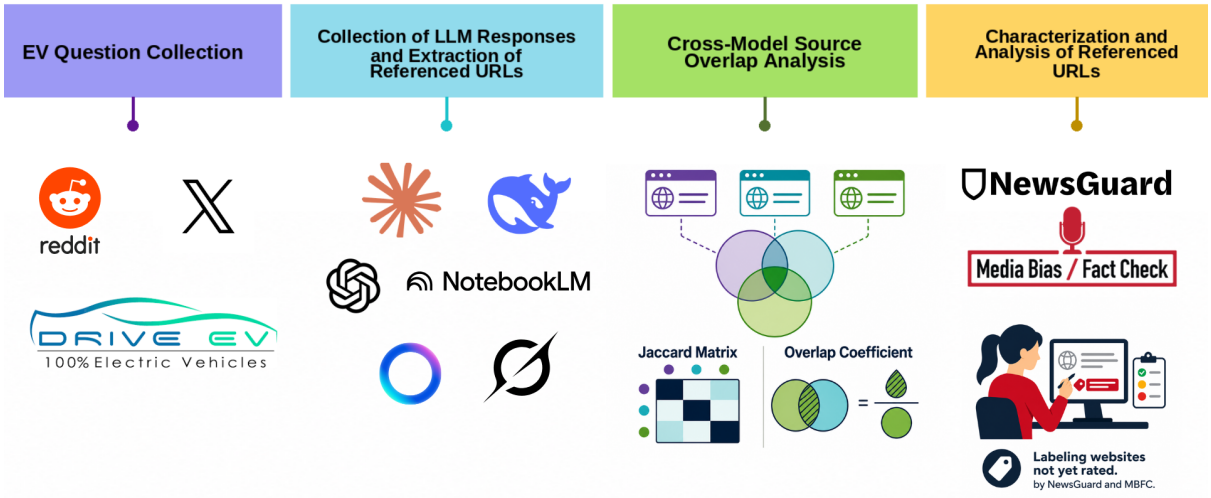


Figure 1: Methodological pipeline

3.1. EV Question Collection

The first phase involves collecting real EV-related questions from multiple online sources, including social media platforms such as Reddit and X, and the informational website DiscoverEV. DiscoverEV is a news portal that provides reviews and guides on EVs[1]. The questions were obtained through a semi-automated extraction procedure. The questions were preserved in their original wording as published in the selected sources.

Some of the questions were collected manually from Reddit and X. On Reddit, the selection focused on the **r/electricvehicles** community; on X, questions were identified using the search bar and the keywords *electric vehicles* and *EV*.

[1]https://www.discoverev.co.uk/

Temporal filters were applied to both platforms, considering the period between **1 January 2025** and **8 January 2026**, to prioritize recent questions aligned with current discussions on the electric vehicle market. For this manual selection, posts with more than one comment were prioritized, since comments represent a strong engagement signal requiring cognitive investment and indicating a high likelihood of discussion and sustained interaction between users [16].

In parallel, automated data collection procedures were adopted. For Reddit, the Python-based extraction relied on RSS feeds and covered the period between **January 1, 2026** and **January 8, 2026**. RSS feeds were used to enable the automated collection of recent Reddit discussions. The extraction relied on the following keywords: *electric vehicle*, *ev*, *electric car*, *tesla*, *nissan leaf*, *mustang mach-e*, *ev charging*, *charging station*, *battery degradation*, and *battery swap*. For DiscoverEV, questions were extracted through web scraping using Python and the Beautiful Soup library. As with the automated Reddit collection, the extracted questions covered the period from **1 to 8 January 2026**.

After collection, the questions were screened by the authors to retain those directly relevant to EVs and representative of plausible information needs among EV owners, prospective buyers and non-specialist users. Technical questions were not excluded per se; those concerning aspects of general relevance to EV use and understanding, such as batteries, charging, maintenance, performance or related technologies, were retained, whereas those focused on highly specialized engineering or very specific configurations were excluded. Similarly, questions involving political or policy-related content were included when they directly addressed issues such as incentives, taxation, regulation or electric mobility infrastructure. By contrast, questions that were predominantly partisan, ideological, electoral or personality-centred were excluded if the reference to EVs was only secondary. This distinction was particularly relevant for X, where discussions involving manufacturers or public figures associated with the sector could be intertwined with broader political controversies.

The inclusion and exclusion criteria adopted during the screening process can be found in Table 1. Following the screening process, we obtained a total of 51 questions: 25 from Reddit; 9 from X; and 17 from DiscoverEV.

Table 1
Question selection criteria

Selection dimension	Inclusion criteria	Exclusion criteria
Topical relevance	Questions whose primary focus concerns electric vehicles, including their adoption, use, cost, reliability, maintenance, charging, performance, environmental impact, infrastructure, market development, or policies directly related to electric mobility.	Items in which electric vehicles are mentioned only incidentally or whose primary topic falls outside electric mobility.
Relevance to non-specialist users	Questions reflecting information needs that may reasonably arise among EV owners, prospective buyers, consumers, or non-specialist users interested in electric mobility.	Questions primarily addressing highly specialized information needs relevant mainly to domain experts and having limited relevance to non-specialist EV users.
Technical specificity	Technical questions concerning components, technologies, maintenance, charging systems, batteries, vehicle range, performance, tyres, Vehicle-to-Grid technologies, or other aspects relevant to the practical use or general understanding of EVs.	Highly specialized engineering- or configuration-specific questions requiring expert-level knowledge of particular components, architectures, circuits, or vehicle configurations and having limited relevance to general EV information needs.

Table 1 – continued from previous page

Selection dimension	Inclusion criteria	Exclusion criteria
Political and policy content	Questions concerning public policies directly related to EVs, including subsidies, taxation, regulation, infrastructure, environmental policies, or other institutional decisions affecting EV adoption and use.	Predominantly partisan, electoral, ideological, or personality-centred questions in which the main information need concerns political actors or political controversies rather than electric vehicles.
Public figures and corporate leadership	References to public figures, executives, or owners of EV manufacturers are retained when the question remains primarily focused on electric vehicles, the company, its products, EV technology, or the EV market.	Questions in which a public or corporate figure becomes the primary subject of discussion because of political, ideological, electoral, or personal issues that are not directly related to electric vehicles.
Brand or model specificity	Questions concerning individual manufacturers or vehicle models when they address issues with broader relevance to EV ownership, reliability, performance, adoption, market development, or technological evolution.	Questions concerning extremely specific problems, configurations, or characteristics of a single vehicle when the information need has very limited relevance beyond that particular case.
Temporal eligibility	Items published within the predefined temporal window established for the corresponding source and collection procedure.	Items published outside the predefined temporal window established for the corresponding source.
Engagement in manual collection	For manually collected social-media questions, engagement signals, particularly the number of comments, were used to prioritize items associated with active user discussion.	Low engagement did not constitute an automatic exclusion criterion, as engagement was used as a prioritization signal rather than as a strict eligibility requirement.

Table 2
Statistics about the collected data. Hostname counts retain the `www.` prefix. Unrated URLs are URL records without a rating from either NewsGuard or MBFC. Unrated Hostnames denotes the number of distinct hostnames represented among these unrated URL records.

Number of LLMs	Total Questions	Total Answers	URL Records	Unrated URL Records	Distinct Hostnames	Unrated Hostnames
7	51	357	1,179	947	906	748

3.2. Collection of LLM Responses and Extraction of Referenced URLs

In the second phase, the selected questions were submitted to seven LLM-based assistants: GPT-5.2, Claude Sonnet 4.5, Grok 4.1, DeepSeek-V3.2, Gemini 3 Flash, Gemini 3-based NotebookLM, and Llama 4-based Meta AI. The questions were submitted through the assistants' respective web interfaces between January 5 and January 12, 2026. Questions collected during the initial phase were submitted between January 5 and 8, while those retrieved subsequently, particularly through RSS feeds, were submitted in the following days. The different collection windows reflect the characteristics of the extraction methods, as RSS feeds primarily provided access to recently published discussions rather than older content. Each question was submitted once to each assistant, preserving its original wording, to approximate a realistic information-seeking scenario in which a user consults an LLM-based assistant about electric vehicles. All questions were submitted through the assistants' official websites using a desktop web browser, without API access. No specific web-search configuration was imposed across the assistants; any automatic web retrieval depended on the functionality and settings of each platform at the time of collection. In the analyzed dataset, URLs were deduplicated separately for each assistant across its 51 responses, retaining one record per distinct URL. When the same URL was cited by different assistants, it was retained once for each assistant. This resulted in 1,179 assistant-specific URL records across 357 responses, corresponding to 906 distinct URL hostnames.

3.3. Cross-Model Source Overlap Analysis

To quantify the degree of overlap in the sources cited by different LLM-based assistants, we consider the set of distinct registrable domains referenced by each assistant across all 51 EV-related questions. The 1,179 collected URL records contain 906 distinct URL hostnames before removing the `www.` prefix. After removing this prefix, the number of distinct hostnames decreases to 811. This normalization consolidates different subdomains belonging to the same registrable domain into a single element. To perform the similarity analysis, purple the 811 distinct hostnames were further consolidated into 772 distinct registrable domains

to assess source overlap at the registrable-domain level, rather than at the hostname level, ensuring that differences in website subdomain structures do not influence the comparison between assistants. For each assistant m, we construct the aggregated domain set $S_m = \{d_1, d_2, \ldots, d_n\}$, counting each domain only once, regardless of the number of URLs associated with it. Therefore, the similarity analysis compares the overall domain sets of the assistants rather than the sources cited in response to individual questions.

We then measure the similarity between models using two complementary set-based metrics, which capture different aspects of source overlap. The Jaccard similarity:

$$J(S_i, S_j) = \frac{|S_i \cap S_j|}{|S_i \cup S_j|}$$

measures the overlap between two sets by capturing the proportion of shared domains relative to the total number of distinct domains cited by the two models. The Overlap coefficient:

$$O(S_i, S_j) = \frac{|S_i \cap S_j|}{\min(|S_i|, |S_j|)}$$

measures the extent to which the smaller set is contained within the larger one, by normalizing the intersection by the size of the smaller set.

Although the Overlap coefficient normalizes the intersection by the size of the smaller set, it remains sensitive to the size of the larger set. To account for this effect, we introduce a size-preserving random baseline. Under this reference model, each assistant independently selects its observed number of distinct registrable domains uniformly at random, without replacement, from a common reference set.

For two assistants i and j, the expected number of shared domains is calculated as

$$\mathrm{E}[|S_i \cap S_j|] = \frac{|S_i| \cdot |S_j|}{N},$$

where $N = 772$ represents the total number of distinct registrable domains in the dataset. Accordingly, the expected Overlap coefficient is

$$\mathrm{E}[O(S_i, S_j)] = \frac{\max(|S_i|, |S_j|)}{N}.$$

This baseline provides a descriptive reference for interpreting the observed overlaps while accounting for differences in domain-set size. It does not represent the actual source-selection mechanisms of the assistants or provide a test of statistical significance.

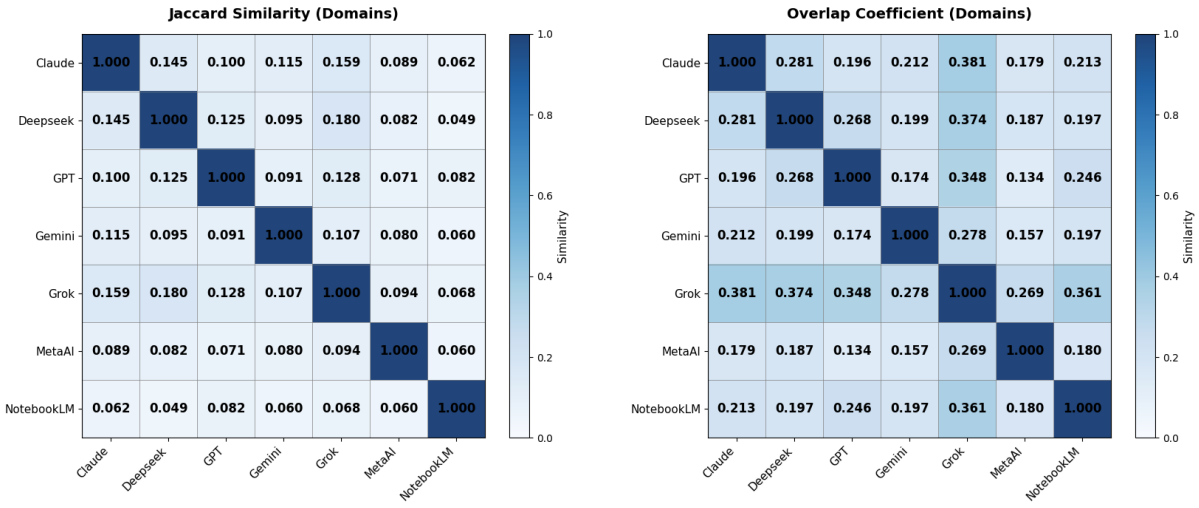
Different units of analysis are adopted to address distinct research objectives. Domain-level comparisons are used to assess source overlap across assistants, counting each website once regardless of the number of associated URLs. Conversely, NewsGuard and MBFC evaluations are summarized at the URL-record level to characterize the collected references, allowing multiple URLs from the same website to contribute separately to the reported counts. Therefore, these analyses provide complementary perspectives rather than directly comparable measures of source diversity and credibility. The descriptive hostname counts reported in Tables 4 and 5 remove the `www.` prefix but otherwise retain subdomains, whereas the similarity analysis further consolidates these into registrable domains.

3.4. Characterization and Analysis of Referenced URLs

To characterize the URLs referenced in LLM responses, we adopt a source-level approach leveraging the evaluations provided by two well-established source-rating services. NewsGuard[2] assesses the reliability of online news outlets through trained journalistic review, while Media Bias/Fact Check [3] evaluates the political bias, credibility, and factual accuracy of news sources. The extracted domains predominantly correspond to English-language

[2]https://www.newsguardtech.com/

[3]https://mediabiasfactcheck.com/



(a) Jaccard similarity matrix

(b) Overlap coefficient matrix

Figure 2: Jaccard similarity matrix and Overlap coefficient matrix

websites based in the United States and the United Kingdom, with a smaller proportion of sources in Italian, French, and German.

For each rating service, we considered the URL records without an associated rating and manually classified the corresponding websites according to their primary function and content. The classification was conducted separately for NewsGuard and MBFC. Records for which a website-type category could not be reliably assigned were excluded from the percentage distributions in Figure 3. For each assistant and rating service, the denominator corresponds to the number of unrated URL records with an assigned website-type category, rather than the total number of unrated records.

We distinguish six website macro-categories based on their primary function and content: Business & commercial services; News & informational resources; Government & civil society; Education, research & culture; Blogs, forums & social platforms; and Other.

4. Results

This section presents the results of our analysis. First, we examine the degree of overlap in the domains cited across the models. Next, we characterize the domains in terms of credibility, factual reporting and political bias. To achieve this, we employ the two source-rating services, NewsGuard and MBFC. For manually classified unrated URL records, we present their distribution across website categories.

4.1. Cross-Model Source Overlap Analysis

As Figure 2a shows, the Jaccard matrix consistently displays low off-diagonal values (from 0.049 to 0.18), indicating limited overlap between the assistants' aggregated sets of cited domains. However, these values are also influenced by differences in domain-set sizes.

The Overlap matrix (Figure 2b) normalizes the intersection by the size of the smaller set but remains influenced by the size of the larger one. It reveals overlap values of up to approximately 0.38. Grok exhibits relatively high overlap with several other assistants, although this pattern must be interpreted cautiously given that Grok has the largest domain set.

To assess the influence of domain-set sizes, we compared the observed Overlap coefficients with a size-preserving random baseline. Across the 21 model pairs, nine observed values exceeded the baseline, while twelve remained below it. As shown in Table 3, only two of the six comparisons involving Grok exceeded the expected Overlap of 0.368: Claude (0.381) and DeepSeek (0.374). Both differences were small, suggesting that Grok's relatively high overlap is largely consistent with expectations based on its domain-set size.

Moreover, when given the same questions through their respective web interfaces, the assistants differed in the number of distinct cited hostnames. As shown in Tables 4 and 5 (column Hostnames), the number of distinct normalized hostnames ranged from 61 for NotebookLM to 289 for Grok.

Table 3
Observed and expected Overlap coefficients for comparisons involving Grok, calculated using registrable-domain sets ($N = 772$).

Assistant	$\lvert S_i \rvert$	$\lvert S_i \cap S_G \rvert$	O_{obs}	$E[O]$
Claude	160	61	0.381	0.368
DeepSeek	195	73	0.374	0.368
GPT	138	48	0.348	0.368
Gemini	151	42	0.278	0.368
Meta AI	134	36	0.269	0.368
NotebookLM	61	22	0.361	0.368

Table 4
NewsGuard evaluations. Scores range over: 100 (High Credibility), 75–99 (Generally Credible), 60–74 (Credible with Exceptions), 40–59 (Proceed with Caution), and 0–39 (Proceed with Maximum Caution). URL records indicates the number of distinct URLs retained for each assistant after deduplication across its 51 responses. Hostnames indicates the number of distinct URL hostnames after removing the `www.` prefix. Unscored denotes URLs lacking a NewsGuard score.

			Credibility Score					
LLM	**URL records**	**Hostnames**	**100**	**75–99**	**60–74**	**40–59**	**0–39**	**Unscored**
Gemini 3-based NotebookLM	72	61	3	2	0	0	0	67
Llama 4-based Meta AI	135	135	2	3	0	0	0	130
Grok 4.1	294	289	24	13	1	1	0	255
Gemini 3 Flash	152	152	7	2	0	0	0	143
DeepSeek-V3.2	200	200	8	8	1	0	0	183
GPT-5.2	156	143	10	6	1	0	0	139
Claude Sonnet 4.5	170	163	11	10	1	1	1	146

We also examined the frequency of normalized URL hostnames within the 1,179 collected URL records. Across all assistants, the most represented normalized URL hostnames were recurrentauto.com (11 URLs), en.wikipedia.org (10), and iea.org (8), followed by rmi.org, edmunds.com, and greencars.com (7 each). At the individual model level, en.wikipedia.org was the most frequent hostname for GPT-5.2 (7 URLs), while recurrentauto.com was the most frequent for NotebookLM (4). For Claude and Grok, the maximum frequency was two URLs per hostname, reached by seven and five hostnames, respectively. Conversely, Meta AI, Gemini, and DeepSeek showed no repeated hostnames within their respective URL sets. These frequencies refer to the collected URL records rather than the total number of citation occurrences in the original responses.

4.2. Characterization of Referenced URLs via NewsGuard

In this section, we characterize the referenced URLs based on the evaluations provided by NewsGuard. NewsGuard evaluates news sites based on a set of criteria that considers both source metadata (e.g., ownership and funding) and article content. News sites receive an aggregated numerical score that determines their credibility level based on the criteria evaluation results.[4]

As shown in Table 4, most of the URLs have no NewsGuard score. For instance, 130 of the 135 URLs cited by Meta AI and 67 of the 72 URLs cited by NotebookLM had no associated NewsGuard score. Overall, 1,063 of the 1,179 URL records (90.2%) have no NewsGuard score. Within our dataset, URLs associated with NewsGuard-rated sources predominantly fall within the two highest score ranges.

4.3. Characterization of Referenced URLs via MBFC

Table 5 characterizes the links in terms of credibility, factual reporting and bias according to MBFC [5]. Consistent with the results obtained using NewsGuard, a clear pattern emerges across all models: 964 of the 1,179 URL records (81.8%) are not evaluated by MBFC. Considering both services jointly, 947 URL records (80.3%) received no rating from either NewsGuard or MBFC, corresponding to 748 distinct URL hostnames before removing the `www.` (Table 2).

[4]https://www.newsguardtech.com/ratings/rating-process-criteria/
[5]Here is a definition of the three metrics: https://mediabiasfactcheck.com/methodology/

Table 5
MBFC evaluations. Credibility levels are categorized as High, Medium, and Low. Factual Reporting levels are: VH (Very High), H (High), MF (Mostly Factual), M (Mixed), and L (Low). Bias levels are categorized as: L (Left), LC (Left-Center), LB (Least Biased/No Bias), RC (Right-Center), R (Right), CP (Conspiracy-Pseudoscience), PS (Pro-Science), and Q (Questionable). Unrated denotes URLs without an MBFC evaluation.

			Credibility			Factual Reporting					Bias								
LLM	URL records	Hostnames	High	Medium	Low	VH	H	MF	M	L	L	LC	LB	RC	R	CP	PS	Q	Unrated
Gemini 3-based NotebookLM	72	61	13	0	0	0	11	2	0	0	1	4	3	0	0	0	5	0	59
Llama 4-based Meta AI	135	135	10	2	0	1	9	2	0	0	1	5	2	0	0	0	4	0	123
Grok 4.1	294	289	53	5	1	3	42	10	3	1	6	19	17	6	2	1	8	0	235
Gemini 3 Flash	152	152	21	5	0	2	12	9	3	0	3	12	5	1	0	0	5	0	126
DeepSeek-V3.2	200	200	23	4	2	1	21	3	2	2	1	9	7	1	2	1	7	1	171
GPT-5.2	156	143	24	4	0	3	19	3	3	0	2	7	10	2	1	0	6	0	128
Claude Sonnet 4.5	170	163	40	7	1	4	29	8	7	0	3	11	18	1	2	0	12	1	122

Most URLs evaluated by MBFC receive high credibility and favorable factual-reporting assessments. Within MBFC's original Bias categories, Left and Left-Center account for 84 URL records, compared with 18 for Right and Right-Center, while 62 are classified as Least Biased. The remaining records are classified as Pro-Science (47), Conspiracy-Pseudoscience (2), or Questionable (2). These categories are retained according to MBFC's classification scheme, although the latter three do not represent positions along the left–right political spectrum. This asymmetry concerns only the evaluated sources and does not establish political bias in the assistants' generated responses.

4.4. Website-Type Characterization of Unrated URL Records.

As shown in Table 2, 947 URL records and 807 distinct URL hostnames were rated by neither NewsGuard nor MBFC. For the qualitative analysis, we classified separately the URL records left unrated by each service: 1,063 records without a NewsGuard score and 964 without an MBFC rating. The two subsets overlap, and both include the 947 records rated by neither service. Website-type categories were assigned whenever the website could be assessed. The *News & informational resources* category, for example, included websites whose primary activity was regularly publishing general or sector-specific news and information.

Figure 3 illustrates the distribution of website categories among manually classified unrated URL records, considering each rating service and LLM-based assistant separately. These URLs were manually classified according to the primary purpose, structure, and content of their corresponding websites. Six macro-categories were considered: *Business & commercial services*, encompassing corporate websites, commercial platforms, automotive manufacturers, and repair shops; *News & informational resources*, including news outlets and informational portals; *Government & civil society*, comprising governmental institutions and non-governmental organizations; *Education, research & culture*, covering educational institutions, research websites, and museums; *Blogs, forums & social platforms*, including blogs, discussion forums, and social media platforms; and *Other*, comprising websites not attributable to the aforementioned categories.

As shown in the figure, Business & commercial services represents the largest share of manually classified unrated URL records, accounting for approximately 39.1% (unweighted mean across the fourteen assistant–service distributions), followed by News & informational resources (32.8%) and Government & civil society (18.4%). Conversely, Education, research & culture is the least represented category (2.6%), followed by Other (3.3%) and Blogs, forums & social platforms (3.9%). These findings highlight the predominance of commercial and informational websites among the sources not covered by the two rating services. The reported percentages describe the composition of the unrated subsets rather than the proportion of websites within each category that remain unrated.

4.5. Takeaways

Drawing on the obtained results, we can address our research questions. **Takeaway RQ1.** The seven LLM-based assistants exhibit differences in the number and composition of their cited-domain sets. Jaccard similarity indicates limited source overlap, while comparison with the random baseline shows that higher Overlap values, particularly for Grok, are partly consistent with expectations based on set size. However, the absence of repeated queries prevents conclusions about the stability of these differences.

Takeaway RQ2. Sources evaluated by NewsGuard and MBFC generally receive favorable credibility assessments. However, MBFC classifications reveal an uneven distribution of political orientations, with Left and

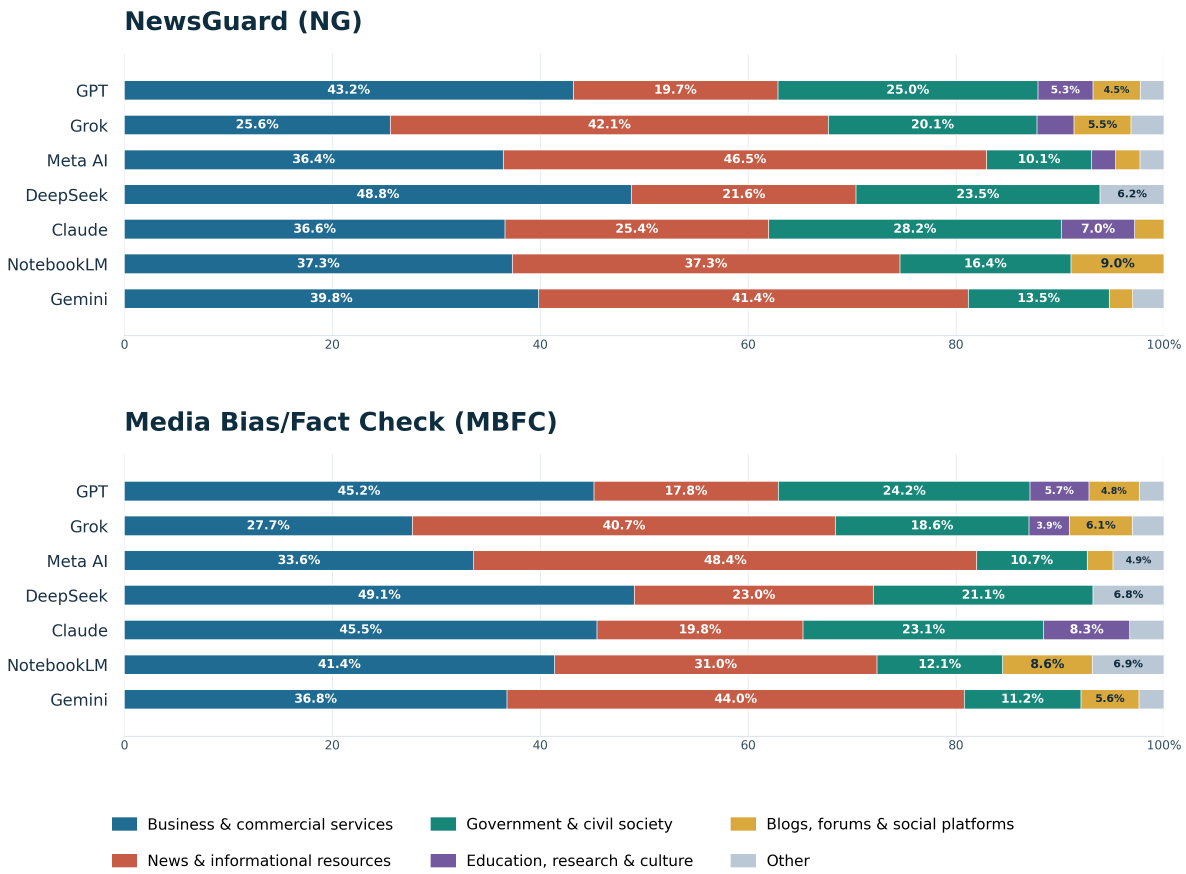


Figure 3: Percentage distribution of website macro-categories among manually classified unrated URL records, shown separately for NewsGuard and MBFC across the seven LLM-based assistants

Left-Center sources (84 URL records) outnumbering Right and Right-Center sources (18). These findings concern only the evaluated subset and do not establish political bias in the assistants' generated responses.

Takeaway RQ3. Most collected sources fall outside the coverage of NewsGuard and MBFC. The manually classified unrated subsets predominantly comprise business and commercial services, followed by news and informational resources, while educational, research, and cultural websites are comparatively less represented. These findings highlight the coverage limitations of existing rating services without implying that unrated sources are necessarily unreliable.

5. Conclusions

This study examined the sources cited by seven LLM-based assistants in response to 51 real-world questions about electric vehicles. The results show differences across assistants in both the composition and size of their cited domain sets. These findings describe the citation patterns observed during the experimental period and should be interpreted in light of several methodological limitations. First, each question was submitted only once to each assistant, so the differences between systems cannot be separated from run-to-run variability or from changes in search results over time. Second, the analysis relies on a limited set of questions, collected through manual and semi-automated procedures and selected through a screening process involving author judgment; a larger or differently constructed set might lead to different results. Furthermore, source overlap was measured using domain sets aggregated across all 51 questions, meaning that two assistants may share a domain even when citing it in response to different questions, rather than the same information need. Extending the question set and repeating the queries over time are natural directions for future work.

Furthermore, the findings indicate that NewsGuard and MBFC cover only a limited proportion of the websites cited in the analyzed EV-related responses. Among the sources evaluated by these services, credibility and factual-reporting ratings were predominantly positive, although political-orientation categories were not evenly distributed, with left-of-center labels more frequent than right-of-center ones. The reliability of unrated sources

was not independently assessed. The qualitative analysis nevertheless revealed a heterogeneous ecosystem of unrated websites, including corporate, industry-specific, and specialized sources, although the manual classification was not accompanied by a measure of inter-annotator agreement.

Within this EV case study, these findings suggest a mismatch between the heterogeneous websites cited by LLM-based assistants and the primary scope of existing news-oriented rating services. Rather than establishing the unreliability of unrated sources, the results highlight the limitations of relying exclusively on existing media-rating frameworks to characterize the sources presented by conversational AI systems.

6. Data Availability

In the interest of transparency and reproducibility, the data used in this study have been made publicly available in a dedicated repository. This repository contains the full set of 51 electric vehicle questions that were considered during the experiment, along with the URLs cited by each of the seven LLM-based assistants that were evaluated. It also contains lists of URLs that were identified as being covered by either Media Bias/Fact Check (MBFC) or NewsGuard during the analysis. The repository is available at: https://github.com/yleniaphd/llm-ev-cited-sources.

7. GenAI Usage Disclosure

During manuscript preparation and revision, the authors used generative AI tools for translation, language editing, and feedback on clarity and internal consistency, including the alignment between research questions, reported analyses, and claims. The tools were not used to generate research data or conduct the empirical analyses. All suggestions were critically evaluated by the authors, who retained full responsibility for the study and the final manuscript.

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